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ORIGINAL PAPER

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Stratigraphy and paleomagnetism of a 2.9-km composite lava section in Eyjafjördur, Northern Iceland: a reconnaissance study

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Abstract Eight mountainside profiles in lava flows south of the fjord Eyjafjördur, Northern Iceland, were sampled for paleomagnetic studies. The sampling was concurrent with pilot stratigraphic mapping of the lava sequences in these and several supplementary profiles. The eight profiles are correlated with minor overlaps so that they form a composite section of 2.9 km thickness, estimated to cover the age range between about 9 and 5 million years ago (Hardarson et al. 1999). Paleomagnetic measurements made on 319 lavas generally yield primary remanence directions of high stability and within-unit consistency. Evidence for at least 17 reversals of the geomagnetic field is seen, as well as numerous field excursions. Frequent clustering of directions in successive lavas indicates that the volcanism was episodic.

Keywords Iceland · Late Miocene · Paleomagnetism · Stratigraphy · Basalt lavas

Introduction

Exposed rocks in Iceland cover the time period from approximately 15 million years (Ma) ago to the present (Hardarson et al. 1997). These rocks are basalts with minor amounts of differentiated products and clastic sediments. The regions where the volcanism was not affected much by glaciations (especially those older than 2–3 Ma) are mostly composed of a fairly regular pile of basalt lavas. Only limited stratigraphic mapping of the lava pile had been carried out before Walker (1959) published his important study on lava sequences in Eastern Iceland. In particular, Walker made use of the presence of petrographically distinct groups of several successive feldsparphyric or olivine-rich lavas: these can sometimes be traced for tens of km within the pile, which predominantly consists of tholeiites.

Measurements of paleomagnetic remanence directions in mountainside profiles within the lava pile were shown to be a promising stratigraphic tool in Iceland by Hospers (1954) and Einarsson (1957). A number of composite sections, each containing hundreds of lava flows, have been mapped since 1964. The methods introduced by Walker have been applied in combination with mapping of paleomagnetic polarity zones, and sometimes K-Ar radiometric dating (Watkins and Walker 1977; McDougall et al. 1977; Kristjansson et al. 1980; Saemundsson et al. 1980; McDougall et al. 1984; Kristjansson et al. 1995, 2003; Kristjansson and Johannesson 1996, 1999; Hardarson et al. 1997). Although these sections have provided new insights into the generation of the lava pile and its subsequent tectonics, they should be considered as being only preliminary reconnaissance surveys. The main reasons for this include: (1) the mapping has been taking place in widely separated cross-sections through the lava pile, and detailed three-dimensional studies are available in only a few cases; (2) the distance between the individual profiles making up the composite sections has often been 10 km or more which may cause localized features to be missed or misinterpreted; (3) areas in the vicinity of the numerous central-volcanoes have generally been avoided, due to complications caused by hydrothermal and tectonic activity.

In the early surveys such as that of McDougall et al. (1977) it was hoped that the rate of buildup of the lava pile was sufficiently constant with time for linear interpolation between K-Ar ages in a section to yield reliable age estimates for geomagnetic chron boundaries and other

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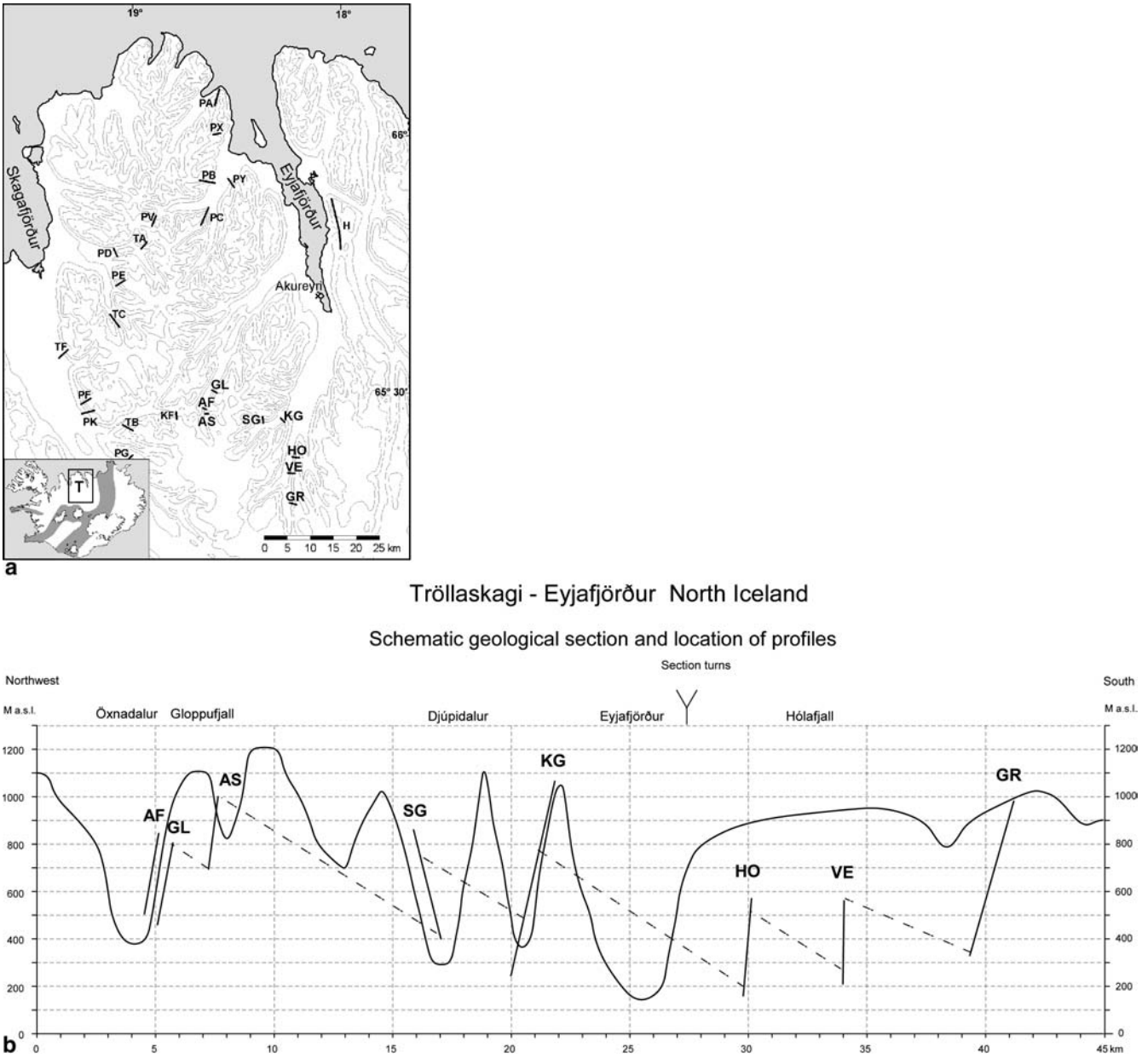


Fig. 1a,b a Index map of the location of eight mountainside profiles GL to GR (*large type*) sampled in the present study, in valleys southwest and south of the town of Akureyri in the Eyjafjördur fjord. The regional tectonic tilt is of the order of 5°, mostly towards the south. The locations of profiles mapped by Saemundsson et al. (1980; see their Fig. 1) farther north and west are shown (*in smaller type*): of these PA, PX, PY, part of PB, PC, most of TA, PE, PK, PF and PG were sampled. A profile KF, which we mapped for purposes of correlation between our section and that of Saemundsson et al. (1980), is also shown. The lengths of the profiles are exaggerated. An 1,800-m thick normal-polarity zone correlated with "Anomaly 5" is found in profiles PC, TA and the lower part of PE. Hebeda et al. (1974) also describe a thick normal-polarity zone (indicated by H) in their composite section east of the fjord. Contour lines are at 250 m intervals. In the *inset map* of Iceland, the active volcanic zones are shown in gray and T indicates the Tröllaskagi peninsula. **b** Simplified schematic of the sampled profiles, projected on an imagined cross-section of the local topography. The cross-section runs southeast from GL and AF in the Öxnadalur valley through AS, SG, and KG, and then south through HO, VE and GR in the main Eyjafjördur valley (Eyjafjardardalur)

events recorded in these sequences. However, later surveys (e.g. Kristjansson et al. 2003) furnished evidence for considerable temporal and spatial variations in the rate of buildup. Processes such as the lateral movement of the major volcanic zones in Iceland have left hiatuses that may reach hundreds of ka; they are not always evident in the field or detectable at the present resolution level of radiometric dates.

One by-product of the stratigraphic mapping projects has been the acquisition of remanence directions and intensities from several thousand lava flows. Due to the long average time between lava flows in a profile (5–10 ka, see e.g. McDougall et al. 1984), the episodic nature

of the volcanism, and the scarcity of radiometric dates, the usefulness of individual lava directions or groups of directions in stratigraphic work is limited. On the other hand, studies on overall statistical properties of these collections of remanence vectors have shed light on some characteristics of the field during the last 15 Ma (Kristjansson 1999, 2002).

Geology of Central Northern Iceland

Previous work

Until 1980, little information was available on the geology of the Eyjafjördur- Skagafjördur region (Fig. 1a), with the exception of Thoroddsen's (1901) geological map of Iceland. A Dutch expedition carried out stratigraphic mapping and paleomagnetic sampling east of Eyjafjördur in the 1960s (Hebeda et al. 1974). They found, among other things, a thick zone of normal polarity lavas in part (around the location indicated by H in Fig. 1a) of their composite stratigraphic section. Radiometric dates obtained by Hebeda et al. indicated, along with dates from the west coast of the fjord (Aronson and Saemundsson 1975), that this zone might correspond to the marine "Anomaly 5" lineation. Both would then correlate with the geomagnetic polarity chron C5n, which covers approximately the interval 9.8–11.0 Ma ago according to current versions of the polarity time-scale (Cande and Kent 1995; Duncan and Helgason 1998). Johannesson (1991) and Johannesson and Saemundsson (1998) indicate that the thick normal-polarity zone in Eyjafjördur extends to about 10 km south of Akureyri at sea-level.

Saemundsson et al. (1980) published laboratory paleomagnetic results from over 450 lavas in a composite stratigraphic section (see profile names in small capital letters in Fig. 1a) in the Tröllaskagi peninsula between Eyjafjördur and Skagafjördur, with 34 K-Ar age determinations. The oldest rocks were found at the north coast of the peninsula, reaching 11–12 Ma in age. The age of an 1,800-m thick normal-polarity zone in their profiles from PC to the lower part of PE agreed reasonably well with that of "Anomaly 5". After "Anomaly 5" time, the buildup of the lava pile south of Skagafjördur was very rapid according to Saemundsson et al. (1980). The youngest dated lavas in their composite section (profile PF) are about 9.0 Ma old.

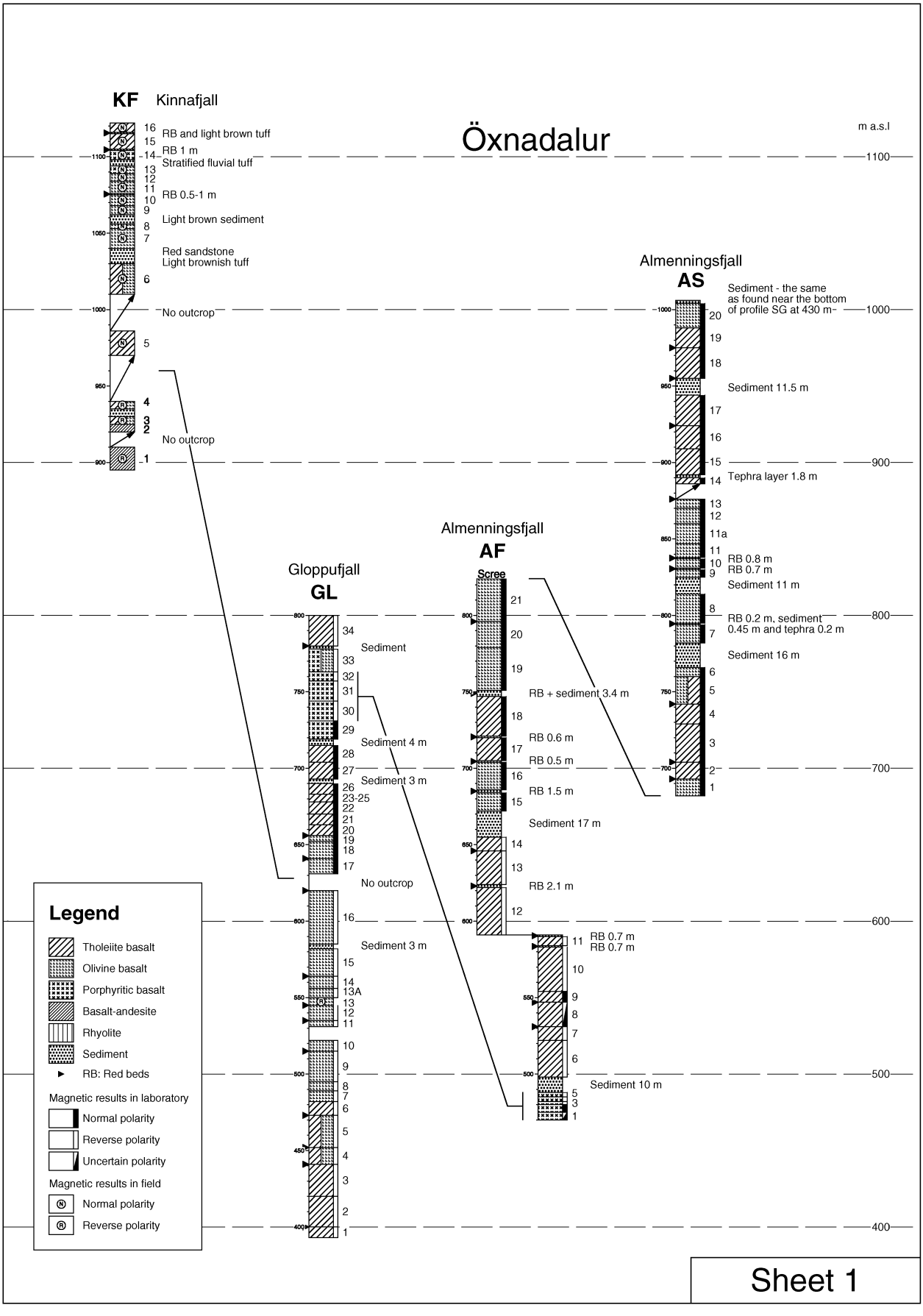
Johannesson (1991) gave a general description of the geology of the Tröllaskagi peninsula. The lava pile has been tilted towards the south in the peninsula but in the area of our composite section there is also an easterly component to the tectonic tilt; Johannesson (1991) relates it to a minor anticline running north-south in between our profiles and those of Saemundsson et al. (1980). Tilt values are often in the range 5–8° at low altitudes, decreasing to 1–3° at mountain tops (1,000–1,400 m). On the basis of field polarity measurements on lavas and the occurrence of cold-climate sediments, Johannesson (1991) and Johannesson and Saemundsson (1998) suggest that the basement age of the highlands 50 km south of Akureyri (i.e. near the scale-bar in Fig. 1a) is 3–3.5 Ma. Johannesson (1991, p. 45) estimates that the thickness of the lava pile from the coast up to there totals 7 km. The only stratigraphic studies so far published in Eyjafjördur have been carried out around Akureyri in connection with geothermal drilling. The largest of these, which covers a 560 km^2 area between locations GL and H in Fig. 1a (Hjartarson and Jonsdottir 1999), did not include laboratory magnetic measurements or radiometric dating.

Geology around the composite section GL to GR

The present pilot project of stratigraphic mapping (Fig. 1a,b) was initiated by one of us (A.G.) and H. Johannesson in several profiles in the Eyjafjördur valleys south of Akureyri around 1990. Profiles in older flows in the Öxnadalur valley southwest of Akureyri were added subsequently. The average distance between profiles is 5 km. The mapping involves the drafting of detailed profile diagrams with information on lava thicknesses, petrography, interbasaltic sediments, and some tectonic features (Fig. 2). The geology of the area is typical for the Tertiary in general: the lava flows that are mostly tholeiites, are around 10 m thick on average. Interbasaltic sediments are usually less than 0.5 m in thickness but occasionally 10 m or more. With one exception mentioned below, the sediments have insufficient lateral extent to be of use in correlation between our profiles. According to Johannesson (1991) and Johannesson and Saemundsson (1998), profile GL lies on the outskirts of a volcanic center named after the Öxnadalur valley and GR lies within another smaller center. However, neither profile appears to have suffered much hydrothermal alteration: remanence intensities are similar in these profiles and the others, and vesicles in lavas in all profiles are generally empty.

Eight of the mapped profiles (Figs. 1, 2), making up a composite section, were selected for paleomagnetic work. The current paper describes the methods involved in the paleomagnetic survey, summarizes the results obtained, and discusses them in the context of a simplified stratigraphical column (Fig. 3). Some paleomagnetic results from this study have already been used along with other similar data from Iceland in research on the overall long-term behavior of the local geomagnetic field (Kristjansson 1999, 2002).

Fig. 2 Detailed drawings of lavas and sediments in the eight profiles sampled for paleomagnetic laboratory studies, as well as in the supplementary profile KF. The drawings cover the valleys Öxnadalur, Djupidalur and Eyjafjardardalur in order of decreasing age. There are three main lava types: lava flows of intermediate or variable character are indicated by *split boxes*. Sediments thinner than 0.5 m are indicated by an *arrowhead* left of the profiles. The altitudes, which were measured by a barometric altimeter, may have errors of up to 20 m. A small fault causing repetition of the flows KG 39–43 as KG 44L-47L, is not shown


Öxnadalur
m a.s.l
KF Kinnafjall
16 RB and light brown tuff
15
RB 1 m
14 Stratified fluvial tuff
13
12
11 RB 0.5-1 m
10
9 Light brown sediment
8
7
Red sandstone
Light brownish tuff
6
No outcrop
5
4
3
2
No outcrop
1
Gloppufjall
GL
34 Sediment
33
32
31
30
29 Sediment 4 m
28
27 Sediment 3 m
26
23-25
22
21
20
19
18
17 No outcrop
16 Sediment 3 m
15
14
13A
13
12
11
10
9
8
7
6
5
4
3
2
1
Almenningsfjall
AF
Scree
21
20
19 RB + sediment 3.4 m
18
RB 0.6 m
17 RB 0.5 m
16
RB 1.5 m
15 Sediment 17 m
14
13
RB 2.1 m
12
11 RB 0.7 m
RB 0.7 m
10
9
8
7
6
Sediment 10 m
5
3
1
Almenningsfjall
AS
Sediment - the same
as found near the bottom
of profile SG at 430 m
20
19
18 Sediment 11.5 m
17
16
15
14 Tephra layer 1.8 m
13
12
11a
11
RB 0.8 m
10 RB 0.7 m
9 Sediment 11 m
8
RB 0.2 m, sediment
0.45 m and tephra 0.2 m
7 Sediment 16 m
6
5
4
3
2
1
1100
1000
900
800
700
600
500
400
Legend
Tholeiite basalt
Olivine basalt
Porphyritic basalt
Basalt-andesite
Rhyolite
Sediment
RB: Red beds
Magnetic results in laboratory
Normal polarity
Reverse polarity
Uncertain polarity
Magnetic results in field
N Normal polarity
R Reverse polarity
Sheet 1

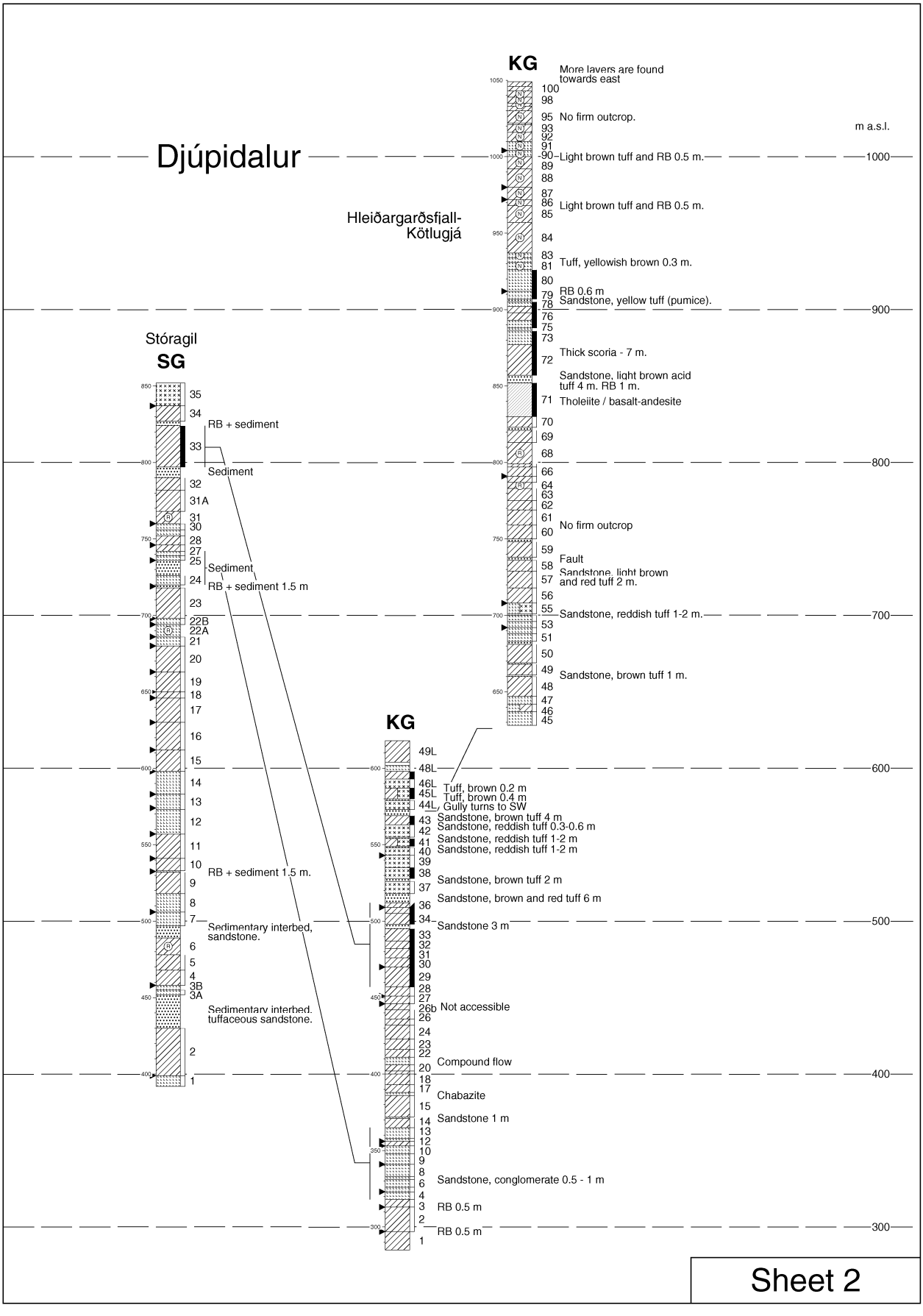


Fig. 2 (continued)

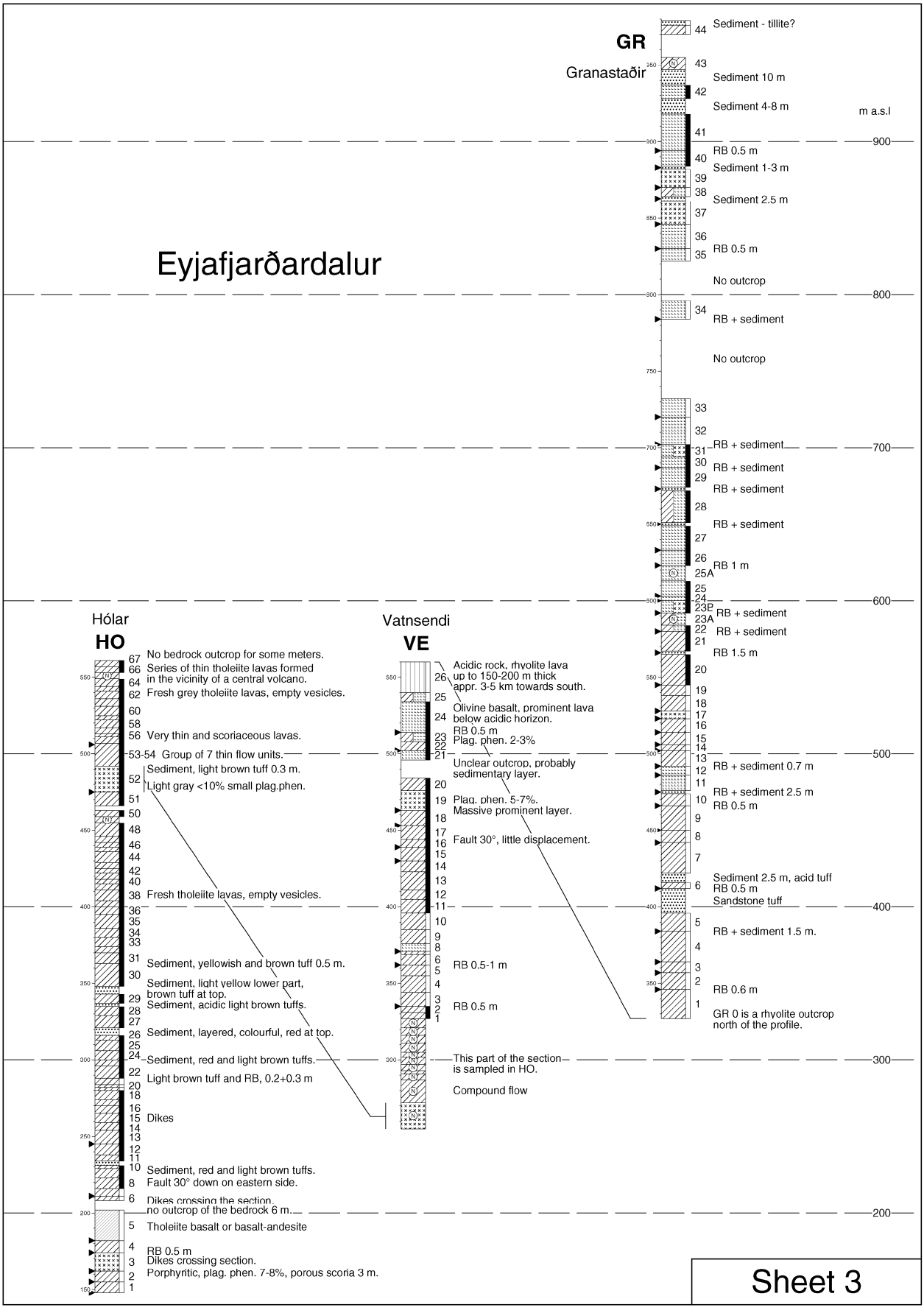


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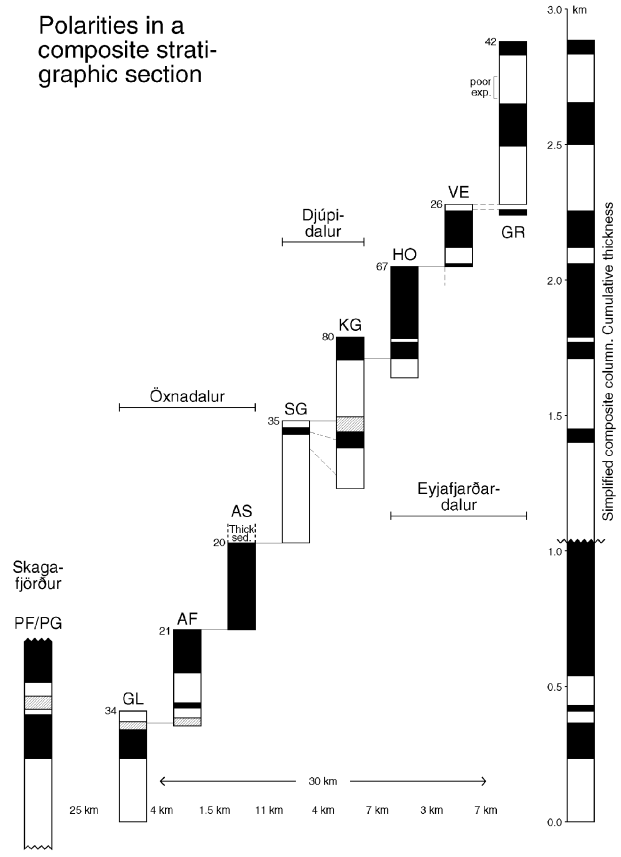


Fig. 3 Geomagnetic polarities in eight profiles of Fig. 2 and Table 1. *Black*: normal polarity; *white*: reverse polarity. Segments where unusual VGP fluctuations occur (Fig. 5) are indicated by *hatching*. Cumulative thickness *on the right*. Correlations between neighboring profiles are explained in the text. *On the left* is the topmost sampled part of the composite section mapped and dated by Saemundsson et al. (1980); we tentatively suggest that the reverse-to-normal transition in their flows PF 35–36 and PG 9–12 is the same as our GL 16–17. In the section of Saemundsson et al. (1980), the flows PG 36 to 41 have fluctuating VGP positions

According to Johannesson and Saemundsson (1998), the top of the thick normal-polarity zone correlated with "Anomaly 5" lies about 10 km northeast of our profile GL in the Öxnadalur valley. We tentatively consider the normal-polarity zone in our profiles GL and KF (Fig. 2) to be the same as one dated by Saemundsson et al. (1980) at about 9.0 Ma at the top of their profile PF. It also occurs in the lower half of their profile PG and in the upper part of their profile TB. In KF and TB, only field measurements of polarity have been made. The age estimate of 9.0 Ma for the upper part of GL agrees with preliminary Ar-Ar dating results from that profile by Hardarson et al. (1999). At the other end of our composite section, preliminary Ar-Ar dates by Hardarson et al. (1999) from its youngest profile GR (Fig. 1b) indicate that it is around 5 Ma old. This is consistent with an age estimate by Johannesson (1991) and Johannesson and Saemundsson (1998) mentioned above. Final results from radiometric age determinations in Eyjafjördur are not yet available but will be published separately along with results from geochemical and isotope-ratio analyses on these basalts.

Description of profile stratigraphy and correlations

The eight profiles sampled (Figs. 1b, 2) to make up a composite polarity column (Fig. 3) are as follows, beginning with the oldest one:

GL Brook in the west side of Mt. Gloppufjall, Öxnadalur valley. Correlation between GL and the next profile AF assumes that a distinct group of lavas GL 30–32 is the same age as the flows AF 1–5. Both groups are feldsparphyric, and large fluctuations in the geomagnetic field (see below) coincided with their emplacement.

AF In the Seldalur tributary valley to Öxnadalur on the west side of Mt. Almenningsfjall, about an hour's walk from the Bakkasel farm ruins.

AS Hillside profile along the valley from AF. Correlation between AF 21 and the bottom flow in AS, which is based on an estimate of the local dip, agrees with magnetic polarities. The profile AS is directly over-lain by coarse indurated sediments that reach at least 100 m in thickness. A few lava flows occur within the sediments but these were not sampled due to difficult access. The sediments above AS are not included in the composite section of Fig. 3.

SG Storagil gully in the Hagardalur tributary to Djupidalur valley. A 20 m layer of tuffaceous sandstone near the base of the profile is considered to correlate with the topmost part of the sediments capping profile AS.

KG Large gully (Kötlugja) in Mt. Hleidargardsfjall, Djupidalur valley. Faults around KG 45 cause repetition of some 60 m of the profile. Correlations with SG are uncertain because the tectonic tilt increases considerably between these profiles. However, allowing for down-dip thickening of the lava series as is commonly observed in Iceland, it is assumed that an olivine-tholeiite lava group SG 24–26 correlates with KG 4–13, and the single normal-polarity flow SG 33 correlates with the polarity zone KG 29–36. The feldsparphyric flow SG 35 might then correspond to the upper part of a porphyritic group KG 37–43.

HO Small stream in Mt. Holafjall near the farm Holar. The lavas are mostly thin tholeiites without significant interbeds, which indicates rapid buildup. Correlation with KG assumes that the R to N transition between KG 70 and 71 is the same as that between HO 7 and 8 and that the short reverse event in HO 19–21 was missed in KG. The correlation is supported by field magnetic measurements on 18 tholeiitic flows overlying KG 80 (not cored and not shown in Fig. 3): all were of normal polarity.

VE Small gully just north of the farm Vatnsendi. Correlation with HO assumes that the thick feldspar-

phyric flow HO 52 is the same as a similar flow seen 70 m below VE 1. We conclude that VE 1 corresponds approximately to the top flow HO 67, and they do indeed yield similar mid-latitude VGPs.

GR Immediately east of the river in Eyjafjardardalur valley, opposite a large gully (Hafrargljufur). The flow VE 26, a thick rhyolite which can be followed part of the way to the profile GR, was expected to be the same unit as a rhyolite GR 0 found 300 m north of GR. However, these two flows have opposite polarities so we assume that there is a minor time gap between GR 0 and 1. There are unexposed segments in the profile above and below flow GR 34, totaling about 70 m.

Paleomagnetism—methods and sample characteristics

Sampling and measurements

The number of basalt lava sites sampled for paleomagnetic measurements was 319. About ten units within the mapped profiles were not sampled due to small or inaccessible outcrops, apparent lack of sediment interbeds, or crumbly texture. Few dikes or other intrusions were noted at the profile locations. At least four 25 mm core samples were collected from each flow, using a portable water-cooled diamond drill. The cores, which were usually taken from the bottom 1–3 m of an outcrop, were spread over a few to several meters laterally. Typical core length is 3–5 cm. The cores were oriented in situ, using a Brunton compass to sight on the Sun or distant geographic objects. One specimen of about 22 mm length was measured from each sample.

All remanence measurements were made at the Science Institute, University of Iceland, using a magnetometer (manufactured by "Institut Dr. Förster" of Reutlingen, Germany) where the specimen to be measured is surrounded by four small fluxgate probes arranged to cancel external field variations. The local magnetic field is compensated by a d-c coil, minimizing noise due to induced magnetization in the specimen. The x-, y- and z-components of the dipole moment of the specimen are each measured in all its eight possible orientations, and averaged. Measurements were carried out on the natural remanence of the specimens, i.e. before alternating field (AF) demagnetization, and immediately after demagnetization at 10, 15 and 20 mT peak fields in a Molspin 2-axis tumbler. Viscous remanence (VRM), which occurs to a very variable extent in the lava flows, was completely removed in over half of the samples by the 10 mT treatment and in most of them by 15 mT. In those cases when the change in direction between the 15 and 20 mT steps exceeded 2–3°, then either (if direction changes appeared to be random) the 20 mT treatment was repeated and the two 20 mT direction results averaged, or (if direction changes showed a systematic trend) a 25 mT step was added for all samples from that flow. 30 mT or higher peak fields, which were applied to a few percent of the collection, led to significant improvement in within-flow agreement of directions in the flow VE 23 (presumably struck by lightning) but to minor or no improvement in most other cases.

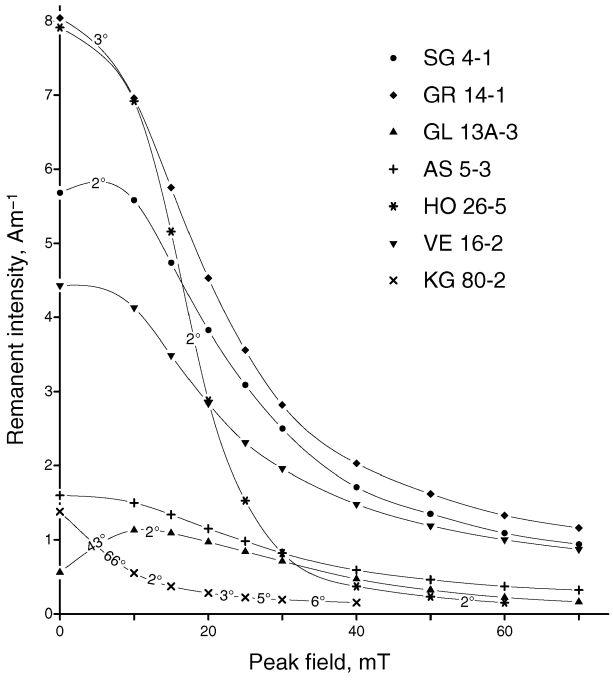


Fig. 4 Extended alternating field demagnetization of seven lava specimens from the present collection. The numbers shown between some data points indicate direction changes, whenever these exceed 1.5°

As the fieldwork took place in five summer seasons (1995–1999), there was opportunity to collect additional samples from some 40 lava flows that had yielded unusual or unsatisfactory results. No meaningful average direction could be obtained from one flow, KG 1, in spite of resampling. Altogether 34 samples from other lavas (mostly in profiles AF and KG) were discarded, due to severe directional instability of remanence or otherwise poor within-lava agreement.

The characteristic remanence

Several specimens, which had reached a stable direction by 20 mT AF treatment, were demagnetized in steps to 70 mT. Their directions in general stayed quite constant (Fig. 4), indicating absence of more than one directionally distinct high-coercivity components. This behavior is also dominant in lavas from other surveys in Iceland (e.g., Kristjansson and Johannesson 1999, Fig. 3). The curve KG 80–2 in Fig. 4 is typical for some occurrences of relatively weak and/or soft remanence as mentioned in the previous paragraph. In such specimens, random magnetization components introduced during or immediately after AF treatment make it impractical to continue the treatment much beyond 20 mT. No certain evidence of

rotational remanence (RRM) could be seen in our samples, but RRM has been noted occasionally at 30 mT AF treatment or beyond in material collected elsewhere in Iceland (e.g., Kristjansson et al. 2003).

The directional results from each lava were averaged, separately for each AF treatment step. Agreement between the directions was usually very good. The average yielding the smallest 95% confidence angle (α_{95}) according to the statistics of Fisher (1953) was selected as the best direction for that flow. If two treatments gave a similar α_{95} within 10%, the average direction from the higher field was used for that lava flow. The rms value of α_{95} in this study is less than 5°, which is very satisfactory.

Table 1 lists all paleomagnetic directions, before and after correction for tectonic tilt as given at the end of the Table. The tilt was measured at or near each profile and is estimated to be correct within 2–3° of arc. Geographic coordinates used in calculating virtual pole positions (VGPs) are given after each profile name. Table 1 also shows the position of the VGP for each lava, and the α_{95} of its average remanence direction. The arithmetic average remanence intensity (J100) of each flow after 10 mT treatment is included in Table 1.

All the direction data from Table 1 were used for statistical calculations below, except for sites KG 44L-47L (duplicate sampling due to a fault), VE 23 (lightning) and GL 23+ (one sample from each of the thin flows GL 23 and 24). The overall remanence intensity in the remaining 312 flows is similar to those previously found in several other large paleomagnetic surveys in the Neogene lava sequences of Iceland: the arithmetic average of J100 is 3.2 A/m and its geometric average is 2.4 A/m.

Median destructive fields for the remanence intensity are very variable; they exceed 20 mT in about 70% of our samples. Magnetic properties in the three main lava types (tholeiites, olivine tholeiites, and feldsparphyric flows) do not exhibit marked differences. Low-field susceptibility measurements were made on one sample from each lava flow with a Bartington MS2 bridge. The susceptibility appears to be uncorrelated with the intensity of primary remanence (see Kristjansson 1999, Fig. 3). Thermal demagnetizations or Curie-point determinations have not been carried out.

Paleomagnetism—results of direction measurements

Serial correlation

The polarities for each sampled lava in Table 1 are shown in the stratigraphic sections for each profile (Fig. 2). In Table 1, one notices that successive paleomagnetic directions are often similar. Although usually only two lavas are involved, examples of larger clusters of directions include GL 21 to 25 (or 28), AF 3–5, AS 5–7, 10–13 and 14–16, SG 25–28, KG 16–22, 23–25 and 47–49, HO 15–17, 43–48, 56–59 and 60–62, GR 22–25, and possibly VE 21–24. The clustering is most noticeable in profile HO, in agreement with the fact that most of the lavas in this profile are thin tholeiites separated only by scoria zones without significant sediment interbeds. Some of these lavas might be more appropriately named "flow units" or "compound flows" (cf. Watkins and Walker 1977; Kristjansson and Johannesson 1999). This serial correlation is quite common in the Miocene to Quaternary lava pile in Iceland (see Kristjansson et al. 1980, Section 5.2; Kristjansson et al. 2003).

McElhinny et al. (1996) proposed a simple method to estimate the number of directionally grouped Late Quaternary lavas in a particular collection where the mean within-site α_{95} was 4°. According to that method, two successive sites were suspected to be close together in time (say <100 years, cf. Mankinen et al. 1985, p 10,408) if the angle between their remanence directions is less than 6°. As the mean α_{95} in our sites is also about 4°, we have applied the test of McElhinny et al. (1996) without modification to the data of Table 1. It yields 65 of the 312 lavas as being suspected of erupting soon after the previous one. However, even in a population of flows erupted at long intervals, a fraction of the directions (of the order of 15 in this case) would be within 6° of the previous one by chance. Kristjansson et al. (2003) have argued that the value of this critical angle should increase with decreasing VGP latitude, but their approach yields similar overall results as that of McElhinny et al. (1996). The directional clusters being small and located in all the profiles of our composite section, they will not significantly affect the mean or the angular standard deviation of the directional distribution of this collection.

Secular variation, excursions and transitions

The remanence directions of Table 1 mostly represent ordinary secular variation between abrupt reversals, as has been noted in many previous surveys in Iceland. "Transitional" lavas with VGP latitudes less than 40° are found at some polarity-zone boundaries: in our survey these include KG 27–28, 71; HO 8, 21; VE 10. More commonly however, a couple of flows will record apparent excursions, in the present case GL 10; SG 12, 19; KG 6, 10–11, 55–58; HO 27–28, 54; VE 18; GR 2. Detailed pole paths such as that reported from Steens Mountain in the Western U.S. (Mankinen et al. 1985) are not often seen in lava sequences: paleomagnetic sampling in Iceland has generally avoided the thick series of flow-units most likely to record such details. Only a single fairly extended low-latitude segment of a polarity transition is known from the Icelandic lava pile, namely the "R3-N3" one first described in the 1950s (see Kristjansson and Sigurgeirsson 1993; Goguitchaichvili et al. 1999).

At two N to R zone boundaries in our composite polarity column, large variations in directions occur through several successive flows. These segments are GL 30–34/AF 1–10 (Fig. 5, middle), and KG 35-43 which is possibly also represented by the flows SG 33–35 (Fig. 5, bottom). Such irregular variations have been noted to occur sporadically in some previous studies elsewhere in Iceland,

Table 1 List of stable remanence directions and virtual pole positions from the present survey

Lava	N	uDec	uInc	Dec	Inc	Lon	Lat	K	Alf	J100	Pol
	GL	GLOPPUFJALL				65.5N	18.6W				
GL 1	4	164.1	-68.9	166	-76	58	-84	686	4	6.23	R
GL 2	4	126.5	-74.0	106	-79	112	-63	446	4	6.40	R
GL 3	4	202.8	-72.1	223	-76	236	-72	442	4	2.86	R
GL 4	4	224.5	-71.8	247	-74	231	-60	1162	3	2.80	R
GL 5	4	154.8	-76.5	149	-83	135	-75	735	3	4.70	R
GL 6	4	148.8	-72.2	143	-78	94	-75	614	4	4.11	R
GL 7	4	200.4	-53.1	206	-57	299	-59	412	5	2.41	R
GL 8	4	195.3	-63.9	204	-69	288	-72	2670	2	6.20	R
GL 9	4	185.0	-65.9	192	-71	304	-79	301	5	6.04	R
GL10	5	286.3	-71.3	298	-67	205	-35	510	3	2.49	RT
GL11	4	176.6	-69.3	183	-75	320	-86	6598	1	1.09	R
GL12	4	196.8	-73.1	213	-77	235	-77	848	3	2.78	R
GL13A	4	189.8	-68.8	200	-74	274	-79	842	3	1.90	R
GL14	4	177.8	-63.3	183	-69	335	-77	789	3	4.06	R
GL15	4	171.9	-51.3	173	-56	352	-61	608	4	2.33	R
GL16	4	212.4	-66.5	223	-69	262	-65	1362	3	4.40	R
GL17	4	352.2	+61.9	355	+67	174	+74	1391	3	2.85	N
GL18	4	33.0	+58.4	40	+61	98	+58	554	4	6.96	N
GL19	4	31.1	+43.9	35	+47	114	+47	365	5	1.57	N
GL20	4	36.0	+59.4	44	+62	93	+57	585	4	4.88	N
GL21	4	1.2	+66.5	6	+71	141	+80	644	4	0.86	N
GL22	4	358.4	+66.5	3	+71	152	+80	459	4	4.16	N
GL23+	2	1.1	+69.8	7	+74	130	+83	412	(12)	0.79	N
GL25	4	5.6	+68.1	11	+72	127	+80	404	4	1.68	N
GL26	4	356.1	+68.5	359	+72	163	+82	256	6	7.33	N
GL27	5	339.0	+64.5	339	+69	210	+73	4092	1	3.60	N
GL28	4	351.6	+63.8	354	+68	177	+75	173	7	2.40	N
GL29	4	262.0	+77.7	243	+78	308	+50	880	3	2.01	N
GL30	4	259.6	-15.3	261	-15	257	-11	98	9	0.52	RT
GL31	8	280.4	-41.2	283	-39	230	-15	195	4	2.28	RT
GL32	4	322.7	-63.6	325	-60	189	-20	1313	3	0.73	RT
GL33	3	203.4	-64.2	210	-67	283	-68	1273	4	0.21	R
GL34	4	165.5	-49.6	166	-54	3	-58	774	3	0.36	R
	AF	ALMENNINGSFJALL				65.4N	18.6W				
AF 1	6	219.9	+24.3	218	+21	304	- 9	189	5	0.93	E
AF 2	6	269.2	+29.4	266	+31	262	+14	51	9	0.44	NT
AF 3	4	315.5	-65.1	320	-60	193	-20	157	7	0.39	RT
AF 4	4	328.9	-59.9	331	-54	186	-13	1095	3	1.57	RT
AF 5	4	332.4	-59.3	334	-53	184	-12	473	4	1.11	RT
AF 6	4	177.0	-70.2	184	-76	303	-87	1858	2	4.53	R
AF 7	6	126.1	-66.2	117	-71	84	-59	93	7	1.17	R
AF 8	7	85.2	+ 4.2	86	+ 3	75	+ 3	62	8	0.49	E
AF 9	5	32.9	+79.8	69	+82	21	+66	76	9	1.87	N
AF10	6	103.2	-77.6	75	-80	126	-55	173	5	1.21	R
AF11	6	163.7	-67.1	165	-73	34	-80	136	6	0.91	R
AF12	5	148.4	-66.4	146	-71	58	-71	261	5	0.68	R
AF13	4	151.7	-63.7	150	-69	44	-70	597	4	1.43	R
AF14	7	175.2	-78.0	185	-83	169	-80	72	7	0.78	R
AF15	4	340.1	+56.0	340	+61	197	+64	775	3	3.49	N
AF16	4	350.3	+82.2	8	+87	344	+71	2126	2	7.59	N
AF17	4	2.7	+67.8	9	+72	130	+81	196	7	5.41	N
AF18	4	6.8	+61.9	12	+66	135	+72	516	4	5.53	N
AF19	4	336.7	+61.2	336	+66	210	+69	690	4	4.85	N
AF20	4	0.9	+46.8	3	+51	157	+57	42	14	2.20	N
AF21	4	19.9	+71.9	32	+75	69	+76	1429	2	1.82	N
	AS	ALMENNINGSFJALL				65.4N	18.6W				
AS 1	4	10.1	+60.5	15	+65	130	+69	1150	3	4.82	N
AS 2	4	318.1	+74.5	309	+79	284	+70	378	5	4.73	N
AS 3	4	323.2	+76.5	314	+81	296	+73	656	4	6.62	N
AS 4	4	352.8	+49.8	354	+55	171	+60	562	4	0.25	N
AS 5	4	10.7	+70.5	20	+75	88	+80	1137	3	1.71	N
AS 6	4	13.6	+65.8	21	+70	110	+74	882	3	2.79	N
AS 7	5	12.4	+67.0	20	+71	107	+76	2488	2	3.05	N
AS 8	4	71.5	+73.7	88	+73	39	+51	899	3	1.07	N
AS 9	4	327.6	+55.6	326	+61	217	+59	1261	3	2.49	N
AS10	4	338.2	+62.7	338	+68	210	+71	533	4	3.11	N
AS11	4	336.1	+62.0	336	+66	210	+69	597	4	2.09	N
AS12	4	347.8	+62.2	349	+66	186	+72	1215	3	7.37	N
AS13	4	339.9	+68.0	340	+72	220	+77	787	3	1.99	N
AS14	4	44.2	+80.6	69	+82	22	+66	507	4	1.78	N
AS15	5	33.1	+77.1	50	+79	39	+71	545	3	1.24	N
AS16	4	17.7	+81.8	44	+84	7	+72	116	9	2.50	N
AS17	4	327.9	+63.6	326	+68	227	+67	910	3	3.29	N
AS18	4	0.8	+44.7	2	+48	158	+54	3224	2	3.18	N
AS19	4	9.5	+67.8	15	+71	117	+78	1010	3	3.06	N
AS20	5	2.6	+76.0	11	+80	20	+84	460	4	5.97	N
	SG	STÓRAGIL				65.4N	18.4W				
SG 1	4	96.6	-80.8	57	-84	143	-58	554	4	2.87	R
SG 2	4	142.0	-60.1	144	-66	48	-65	413	5	0.96	R
SG 3A	4	185.7	-71.3	202	-74	265	-79	710	4	3.24	R
SG 4	4	176.7	-70.6	191	-75	287	-84	940	3	2.97	R
SG 5	4	206.6	-71.2	225	-72	252	-68	2625	6	0.52	R
SG 7	4	154.6	-78.1	172	-83	153	-78	4025	1	3.72	R
SG 8	4	155.1	-61.5	160	-67	25	-71	272	6	2.53	R
SG 9	4	212.1	-69.2	228	-70	255	-64	1577	2	2.63	R
SG10	4	229.6	-65.8	242	-65	250	-52	344	5	5.23	R
SG11	4	124.9	-59.6	123	-66	69	-55	3445	2	1.62	R
SG12	4	57.3	-50.9	50	-52	120	-15	450	4	0.70	RT
SG13	5	179.4	-55.9	185	-59	332	-65	864	3	3.43	R
SG14	4	204.6	-72.2	220	-73	252	-71	1766	2	1.01	R
SG15	4	174.4	-65.4	186	-69	326	-76	1432	2	0.85	R
SG16	4	94.2	-84.4	33	-86	154	-59	1644	2	1.45	R
SG17	4	154.3	-72.1	161	-77	75	-82	1232	3	5.59	R
SG18	4	183.6	-70.1	196	-73	287	-80	3148	2	1.95	R
SG19	4	61.4	-62.7	52	-64	123	-28	395	5	0.22	RT
SG21	4	160.9	-63.6	166	-68	14	-74	346	5	4.24	R
SG22B	4	254.2	-78.6	271	-76	212	-53	302	5	1.40	R
SG23	5	263.8	-82.0	283	-78	201	-53	1352	2	1.52	R
SG24	4	135.4	-66.5	136	-72	69	-67	763	3	1.12	R
SG25	4	183.1	-74.2	196	-77	253	-83	937	3	4.13	R
SG26	4	189.3	-74.9	203	-77	244	-80	448	4	4.36	R
SG27	4	184.4	-71.2	195	-73	286	-81	473	4	7.18	R
SG28	4	185.5	-72.9	197	-75	268	-82	575	4	5.13	R
SG29	4	164.5	-70.8	171	-74	24	-84	149	8	4.25	R
SG30	4	158.8	-72.4	165	-76	61	-83	2596	2	7.52	R
SG31A	4	192.9	-79.1	214	-81	209	-76	241	6	11.72	R
SG32	4	138.5	-59.1	139	-63	49	-60	215	6	1.17	R
SG33	4	357.3	+58.7	2	+62	157	+67	487	4	0.97	N
SG34	3	350.8	-68.8	346	-65	172	-24	2980	2	0.83	RT
SG35	4	98.5	-67.3	92	-70	101	-48	305	5	0.56	R
	KG	KÖTLUGJÁ				65.45N	18.3W				
KG 1	8	Unstable, scattered directions								0.54	(?)
KG 2	4	160.3	-51.5	160	-63	18	-66	850	3	1.76	R
KG 3	6	150.3	-63.4	144	-74	71	-73	10	10	1.87	R
KG 4	6	164.5	-65.6	168	-76	66	-85	56	9	4.25	R
KG 5	4	171.2	-38.6	174	-49	351	-55	573	4	2.39	R
KG 6	7	206.5	- 7.4	208	-15	309	-29	24	12	0.24	RT
KG 7	4	167.5	-63.8	173	-75	23	-85	436	4	2.80	R
KG 8	4	172.4	-66.2	182	-77	287	-89	1333	3	2.57	R
KG 9	4	201.0	-67.2	226	-74	245	-69	613	4	1.53	R
KG10	6	294.8	-75.9	314	-67	195	-30	159	5	0.22	RT
KG11	6	277.5	-53.3	288	-48	224	-19	91	7	0.57	RT
KG12	4	168.4	-79.0	221	-88	168	-68	714	3	0.90	R
KG13	8	184.3	-79.0	249	-86	184	-67	251	4	1.88	R
KG14	4	162.3	-65.3	164	-75	55	-82	324	5	3.06	R
KG15	4	188.2	-53.9	197	-62	310	-66	1008	3	2.69	R
KG16	4	172.3	-71.9	186	-81	175	-82	1134	3	4.89	R
KG17	5	172.0	-70.5	184	-80	175	-85	85	8	3.23	R
KG18	4	178.8	-69.9	195	-79	217	-83	531	4	2.99	R
KG19	4	166.8	-72.7	176	-83	155	-80	946	3	1.80	R
KG20	5	160.6	-71.4	161	-81	130	-80	611	4	2.43	R
KG21	4	165.1	-69.9	169	-79	112	-85	162	7	2.42	R
KG22	4	169.6	-66.1	175	-75	14	-86	140	8	2.06	R
KG23	5	160.7	-81.6	318	-89	164	-64	169	5	2.04	R
KG24	4	160.4	-78.2	162	-87	156	-71	304	5	6.14	R
KG25	4	179.6	-75.4	205	-83	185	-76	269	6	2.62	R
KG26	6	133.0	-58.1	124	-66	68	-56	49	10	1.62	R
KG27	4	196.5	-10.9	198	-18	321	-33	534	4	0.58	RT
KG28	4	196.0	-15.9	198	-23	320	-35	1000	3	0.97	RT
KG29	4	9.3	+64.5	22	+72	100	+76	878	3	1.23	N
KG30	4	318.9	+72.7	299	+81	294	+68	779	3	4.00	N
KG31	5	346.4	+54.8	348	+63	185	+68	253	5	2.29	N
KG32	5	351.1	+54.2	354	+62	173	+67	289	5	4.55	N
KG33	4	44.6	+70.0	67	+72	54	+58	437	4	1.87	N
KG34	4	314.9	+66.9	303	+74	266	+64	155	7	0.60	N
KG35	7	329.6	+34.1	328	+42	203	+44	80	7	0.70	N
KG36	6	342.4	-41.1	342	-33	179	+ 5	46	10	0.43	E
KG37	4	181.1	-65.0	190	-72	307	-81	556	4	0.56	R
KG38	4	260.1	+55.3	248	+56	286	+25	682	4	0.69	NT
KG39	4	190.0	-67.7	204	-74	265	-78	1516	3	0.53	R
KG40	4	145.3	-75.0	130	-82	124	-71	1059	3	2.63	R
KG41	7	64.8	+75.8	92	+75	34	+53	309	3	1.05	N
KG42	4	138.7	-20.5	138	-27	32	-31	218	6	0.86	RT
KG43	8	338.3	+25.6	338	+33	189	+40	21	12	0.60	N
KG44L	5	170.0	-65.3	174	-72	5	-81	59	10	0.75	R
KG45L	4	46.5	+73.3	71	+74	46	+60	1567	2	1.58	N
KG46L	4	140.7	-16.1	140	-23	28	-30	116	9	0.65	RT
KG47L	6	340.2	+22.7	340	+30	186	+39	24	14	0.84	NT
KG45	4	185.1	-49.6	189	-56	326	-60	57	12	1.58	R
KG46	5	173.3	-62.4	177	-69	349	-77	1950	2	6.20	R
KG47	4	197.2	-62.3	207	-67	286	-69	1847	2	0.69	R
KG48	4	203.3	-60.1	213	-65	282	-64	1602	2	1.25	R
KG49	4	213.4	-59.6	224	-63	270	-58	1211	3	1.64	R
KG50	4	241.2	-62.5	255	-63	242	-45	201	7	1.61	R
KG51	4	209.6	-67.7	223	-71	257	-67	603	4	1.13	R
KG52	4	202.1	-66.9	214	-71	267	-71	486	4	6.10	R
KG53	4	174.2	-62.7	178	-69	347	-76	498	4	3.62	R
KG54	4	155.4	-78.2	151	-84	140	-75	937	3	3.12	R
KG55	4	23.4	+ 4.2	24	+ 9	135	+27	105	9	0.10	NT
KG56	4	207.4	-37.0	211	-41	301	-44	97	9	0.49	R
KG57	4	201.5	-41.2	205	-46	307	-48	426	5	0.54	R
KG58	4	209.5	+70.1	199	+66	328	+24	1096	3	3.03	NT
KG59	4	182.8	-61.9	188	-67	322	-74	382	5	1.87	R
KG60	4	125.7	-72.0	112	-76	102	-62	370	5	2.60	R
KG61	4	207.9	-70.9	221	-74	249	-71	1372	3	5.62	R
KG62	4	165.4	-71.3	167	-76	62	-84	402	5	3.35	R
KG63	4	138.6	-71.9	131	-76	91	-70	609	4	2.37	R
KG65	4	188.3	-68.6	196	-73	288	-80	336	5	3.57	R
KG66	4	152.1	-63.1	150	-68	43	-69	941	3	4.69	R
KG67	4	141.7	-78.5	129	-83	127	-71	736	3	8.20	R
KG69	5	168.2	-66.5	170	-71	13	-80	151	6	6.52	R
KG70	5	205.2	-85.5	280	-86	178	-63	397	4	4.14	R
KG71	3	209.2	+57.3	205	+55	321	+12	153	10	0.41	NT
KG72	5	312.3	+71.9	305	+75	270	+67	261	5	4.09	N
KG73	4	266.3	+75.4	251	+76	302	+49	5127	1	3.98	N
KG75	4	342.5	+46.5	343	+51	187	+54	438	4	1.42	N
KG77	4	328.1	+81.6	318	+85	322	+71	254	6	1.75	N
KG78	4	313.6	+70.2	308	+74	263	+66	945	3	1.44	N
KG79	4	337.0	+59.0	337	+63	205	+65	364	5	2.04	N
KG80	4	307.9	+73.8	299	+77	280	+65	279	6	1.08	N
	HO	HÓLAR				65.4N	18.2W				
HO 1	5	202.5	-53.7	212	-56	293	-55	765	3	1.86	R
HO 2	4	201.3	-53.2	211	-55	295	-56	8911	1	16.47	R
HO 3	4	218.9	-68.9	237	-68	250	-59	901	3	1.57	R
HO 4	4	192.4	-59.1	204	-62	299	-65	428	4	1.90	R

Table 1 (continued)

Lava	N	uDec	uInc	Dec	Inc	Lon	Lat	K	Alf	J100	Pol
HO 5	6	195.2	-49.4	203	-53	307	-55	687	3	2.01	R
HO 6	4	152.2	-51.1	155	-58	21	-59	1563	2	1.56	R
HO 7	4	135.9	-51.6	136	-59	48	-54	487	4	1.16	R
HO 8	7	299.3	+ 6.8	299	+13	228	+18	74	7	0.46	NT
HO 9	6	325.0	+61.3	328	+68	227	+69	44	10	1.22	N
HO10	5	345.4	+52.6	351	+59	177	+63	537	3	1.85	N
HO11	5	326.8	+64.9	331	+72	233	+74	458	4	3.94	N
HO12	4	355.6	+77.0	26	+81	22	+78	213	6	1.66	N
HO13	4	320.6	+56.1	322	+63	225	+61	1304	3	5.49	N
HO14	4	321.8	+57.9	323	+65	226	+63	825	3	4.90	N
HO15	4	338.4	+74.1	355	+80	325	+84	1172	3	3.37	N
HO16	4	327.9	+75.5	339	+82	314	+79	421	5	7.26	N
HO17	4	339.3	+72.8	351	+78	280	+86	332	5	1.80	N
HO18	4	14.7	+47.0	21	+50	131	+53	392	5	2.03	N
HO19	4	157.9	-72.0	169	-77	82	-85	181	4	0.41	R
HO20	4	130.8	-40.4	130	-46	46	-41	97	9	0.55	R
HO21	8	134.0	-32.3	134	-38	39	-37	441	3	0.37	RT
HO22	4	319.9	+83.1	349	+89	341	+67	716	3	2.68	N
HO23	5	319.0	+67.8	320	+74	253	+71	355	4	3.37	N
HO24	4	352.3	+75.4	12	+79	24	+84	580	4	6.78	N
HO25	4	350.2	+51.0	355	+56	170	+61	1471	2	3.17	N
HO26	6	357.6	+45.9	2	+50	158	+56	70	8	4.14	N
HO27	6	249.9	+53.6	242	+56	292	+22	120	6	0.37	NT
HO28	7	275.4	+50.5	270	+55	268	+32	126	5	1.45	NT
HO29	4	346.9	+56.6	353	+62	176	+67	406	5	2.44	N
HO30	4	23.5	+73.4	44	+75	63	+70	1501	2	4.24	N
HO31	4	16.7	+70.0	33	+72	84	+72	343	5	6.44	N
HO32	4	46.7	+79.3	76	+78	35	+61	689	4	3.12	N
HO33	4	2.1	+61.1	10	+64	142	+70	855	3	4.17	N
HO34	4	71.8	+70.0	83	+67	51	+47	1440	2	2.99	N
HO35	4	317.6	+67.5	318	+72	250	+69	284	6	3.05	N
HO36	4	328.3	+68.2	332	+73	237	+75	989	3	3.40	N
HO37	4	350.8	+69.8	1	+74	158	+84	3657	1	9.46	N
HO38	4	351.2	+66.4	359	+70	164	+79	351	4	3.98	N
HO39	4	286.5	+60.1	282	+64	266	+46	892	3	2.91	N
HO40	4	297.7	+58.5	295	+63	254	+50	469	4	4.16	N
HO41	4	324.2	+61.8	326	+67	226	+66	2856	2	2.69	N
HO42	5	354.4	+63.6	2	+67	157	+75	537	3	3.62	N
HO43	4	337.4	+57.6	341	+62	197	+66	859	3	3.97	N
HO44	4	346.3	+60.3	352	+65	179	+70	2133	2	9.10	N
HO45	4	337.4	+53.9	340	+58	195	+62	2210	2	4.38	N
HO46	4	337.2	+52.4	340	+57	194	+60	420	5	5.80	N
HO47	4	340.8	+53.0	344	+57	188	+61	619	4	5.29	N
HO48	4	330.8	+52.4	333	+57	205	+58	63	12	1.06	N
HO50	4	322.0	+63.4	323	+67	231	+66	4096	1	6.77	N
HO51	4	297.0	+71.7	292	+76	278	+62	661	4	2.66	N
HO52	4	323.5	+63.2	325	+67	228	+66	836	3	3.16	N
HO54	4	259.1	+65.7	251	+68	292	+38	207	6	0.46	NT
HO55	4	52.4	+74.4	66	+73	52	+60	1564	2	3.44	N
HO56	4	38.7	+63.5	47	+64	87	+58	772	3	4.27	N
HO57	4	39.1	+64.4	48	+65	85	+59	443	4	4.54	N
HO58	4	31.3	+62.7	39	+63	97	+61	409	5	8.80	N
HO59	4	31.8	+64.1	40	+65	93	+62	296	5	3.66	N
HO60	4	346.6	+49.2	349	+53	178	+57	1011	3	2.93	N
HO61	4	345.5	+54.4	349	+58	181	+62	494	4	3.33	N
HO62	4	347.9	+53.0	351	+56	177	+61	277	6	4.61	N
HO63	4	334.3	+77.1	342	+81	306	+81	974	3	5.13	N
HO64	4	343.9	+78.3	357	+82	336	+82	811	3	2.73	N
HO66	5	321.3	+74.6	323	+79	278	+76	204	5	3.09	N
HO67	5	54.3	+64.7	62	+64	72	+52	454	4	2.64	N
	VE		VATNSENDI			65.3N	18.2W				
VE 1	4	69.4	+64.8	78	+62	59	+44	349	5	3.16	N
VE 2	4	62.9	+82.5	90	+80	22	+59	731	3	2.31	N
VE 3	4	182.8	-55.1	189	-58	327	-63	1361	3	4.11	R
VE 4	4	177.9	-57.9	184	-61	334	-67	322	5	1.97	R
VE 5	4	122.6	-70.2	119	-75	93	-64	933	3	2.64	R
VE 6	4	138.3	-58.1	139	-63	49	-60	602	4	1.37	R
VE 7	4	161.6	-49.6	165	-54	6	-58	726	3	2.84	R
VE 8	4	185.9	-68.5	197	-71	292	-77	446	4	0.87	R
VE 9	4	232.6	-59.1	241	-58	259	-46	581	4	1.99	R
VE10	3	237.7	-49.3	243	-48	263	-37	4245	2	1.94	RT
VE11	4	350.5	+54.1	355	+58	170	+63	480	4	3.92	N
VE12	4	11.0	+61.3	19	+64	124	+68	70	11	1.14	N
VE13	4	340.2	+68.6	347	+73	210	+81	1201	3	4.80	N
VE14	4	21.9	+59.3	30	+61	111	+61	994	3	1.48	N
VE15	4	326.3	+70.1	330	+75	251	+76	4725	1	8.30	N
VE16	4	329.0	+61.0	332	+66	217	+67	291	5	2.37	N
VE17	4	338.9	+70.8	345	+74	226	+82	1921	2	3.39	N
VE18	4	251.0	+63.7	243	+65	296	+33	1732	2	3.23	NT
VE19	4	357.5	+68.9	6	+72	142	+81	901	3	5.50	N
VE20	4	356.1	+70.2	5	+73	141	+83	979	3	7.09	N
VE21	4	342.7	+62.0	347	+66	190	+71	1076	3	4.35	N
VE22	4	343.6	+61.0	348	+64	187	+70	2425	2	2.83	N
VE23	4	305.7	+64.8	304	+69	254	+59	14	21	(15.53)	N
VE24	4	333.2	+66.2	337	+70	218	+74	6876	2	7.24	N
VE25	4	200.8	-74.5	215	-76	244	-75	256	6	7.36	R
VE26	4	210.1	-82.1	239	-82	201	-69	120	8	0.07	R
	GR		GRANASTAÐIR			65.3N	18.2W				
GR 0	5	28.4	+68.0	38	+69	87	+67	1021	2	2.29	N
GR 1	4	149.5	-58.7	151	-63	32	-63	246	6	5.26	R
GR 2	6	137.8	-19.8	138	-24	31	-30	869	2	1.07	R
GR 3	8	181.3	-33.7	183	-36	337	-45	82	6	2.94	R
GR 4	5	180.5	-35.8	183	-38	338	-46	241	5	1.92	R
GR 5	5	190.8	-72.7	203	-75	263	-79	4182	1	4.55	R
GR 6	4	154.5	-85.7	223	-89	167	-67	1118	3	3.16	R
GR 7	5	178.2	-67.7	186	-70	324	-79	178	6	0.92	R
GR 8	5	180.2	-73.6	192	-76	264	-85	4591	1	1.91	R
GR 9	4	145.6	-68.7	148	-73	61	-73	1312	3	7.23	R
GR10	4	131.6	-79.4	130	-83	129	-71	1702	2	1.64	R
GR11	4	136.4	-60.7	137	-65	54	-60	2433	2	11.83	R
GR12	5	175.5	-43.2	178	-46	344	-52	254	5	1.99	R
GR13	4	157.6	-64.5	161	-68	25	-73	1485	2	4.07	R
GR14	4	157.6	-75.0	165	-78	103	-84	1107	3	7.11	R
GR15	4	214.2	-73.2	228	-74	244	-68	2236	2	4.22	R
GR16	4	256.8	-75.3	268	-73	219	-51	2745	2	3.46	R
GR17	4	149.1	-65.1	152	-69	43	-71	1342	3	3.31	R
GR18	4	123.9	-85.9	47	-89	159	-64	1333	3	6.65	R
GR19	4	167.1	-65.1	172	-68	1	-76	210	5	4.63	N
GR20	4	318.5	+68.7	319	+72	250	+70	580	4	2.79	N
GR21	4	1.5	+81.1	27	+83	7	+76	898	3	4.40	N
GR22	4	4.4	+68.8	13	+71	121	+78	2573	2	8.72	N
GR23B	4	7.3	+65.6	15	+68	126	+74	2765	2	8.47	N
GR24	4	354.9	+66.4	2	+69	157	+78	764	3	3.92	N
GR25	4	347.0	+64.6	352	+68	181	+75	780	3	8.38	N
GR26	6	265.4	+81.1	240	+83	320	+56	65	8	4.65	N
GR27	4	18.0	+65.5	26	+67	107	+69	450	4	7.08	N
GR28	4	14.4	+64.3	22	+66	116	+70	266	6	2.66	N
GR29	4	0.0	+85.1	54	+87	357	+69	2597	2	1.18	N
GR30	4	347.5	+78.3	2	+81	346	+82	492	4	6.09	N
GR31	4	63.3	+79.6	81	+78	32	+59	317	5	2.91	N
GR32	5	170.1	-36.6	172	-40	353	-47	634	5	1.13	R
GR33	4	134.8	-51.4	135	-54	45	-50	552	4	4.10	R
GR34	4	139.6	-50.4	140	-53	39	-51	1670	2	4.68	R
GR35	4	206.5	-39.2	209	-40	304	-43	569	4	0.62	R
GR36	4	169.9	-57.1	173	-59	354	-65	911	3	5.64	R
GR37	4	148.4	-66.3	151	-69	45	-71	683	4	10.67	R
GR38	5	157.1	-81.7	171	-84	154	-76	996	2	6.25	R
GR39	4	129.5	-78.7	129	-82	120	-71	5839	1	2.98	R
GR40	4	6.6	+69.2	14	+71	122	+78	389	5	6.67	N
GR41	4	354.6	+46.0	357	+48	166	+54	536	4	3.46	N
GR42	5	8.0	+66.2	14	+67	129	+74	1434	2	10.37	N

Tectonic tilt (Deg.) and Down-Dip Directions
The tilt decreases upwards in most profiles

GL:	7 TO 4, SSE	AF:	6 TO 5, SSE
AS:	5 TO 4, SSE	SG:	6 TO 4, SE
KG:	11 TO 4, SSE	HO:	6, SE
VE:	5 TO 4, SE	GR:	4 TO 3, SE

N=number of samples used in averaging. uDec, uInc, Dec, Inc=declination (east of north) and inclination (down) of the mean direction, respectively before and after tectonic tilt correction (see at the end of the Table). Lon, Lat=east longitude and north latitude of the corresponding virtual pole. k, Alf=Fisher's (1953) precision parameter and α_{95} for the mean direction. J100=arithmetic mean remanence intensity after 10 mT AF treatment, A/m. Pol=polarity (N=normal, R=reverse, with T and E indicating VGP latitudes numerically less than 40° and 10° respectively)

e.g. in 7–13 Ma old lavas in NW-Iceland (McDougall et al. 1984; Kristjansson and Johannesson 1996), and in the presumed Gilbert-Gauss transition in Mt. Skardsheidi, SW-Iceland (Kristjansson and Sigurgeirsson 1993; Kristjansson and Gudmundsson 2001). One mechanism that could produce irregularly varying directions like those in flows KG 35–43, would be the intrusion of a dike close to the sampling sites (e.g. Duncan and Helgason 1998). However, such effects would be very localized. The profile KG happens to be crossed by a fault, causing repetition. Four thin lava flows which we sampled north of the fault, KG 44L-47L, are seen to yield scattered directions very similar (allowing for experimental errors) to those in four of the sites KG 39–43 sampled some tens of meters away south of the fault. It is difficult to imagine this agreement in directions as resulting from an artifact.

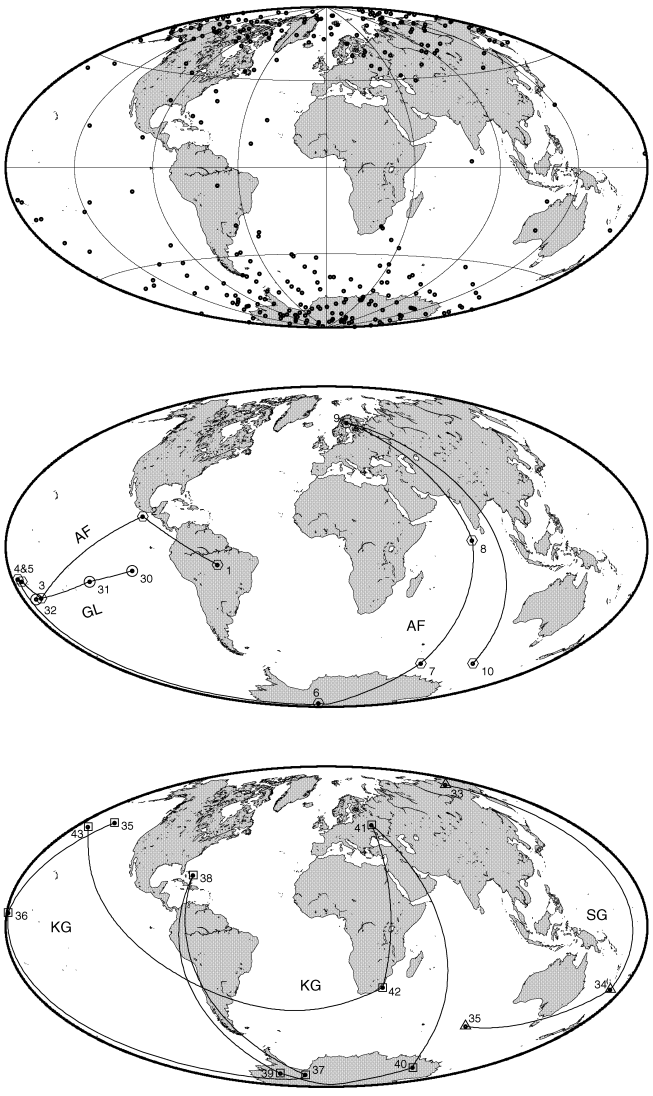


Fig. 5 *Top*: All virtual geomagnetic pole positions from the present study. *Middle and bottom*: apparent irregular variations of the geomagnetic field, recorded in profiles GL and AF, and in profiles SG and KG respectively. The sites KG 35 and 43 are petrographically different, although both are tholeiites. Data from Table 1

Overall averages

The mean field directions from the flows in Table 1 are given in Table 2 along with statistical parameters on the assumption that the directions and VGPs conform to Fisher's (1953) distribution. It should, however, be kept in mind that Icelandic VGPs do not fit that distribution: using a data base of over 3,500 lava flows including those from profiles KG, HO,VE and GR of the present collection, Kristjansson (1999) confirmed previous suggestions that the VGPs are better described by a Bingham distribution or a combination of 90% Fisher-distributed poles and 10% random poles. In Table 2, we list both means from all the data and from units with VGP latitudes exceeding 45°N or S. In the latter, it is seen that the difference between the mean field for normal-polarity flows and that for reverse flows is 2°, which is not significant at the 90% level according to F-ratio tests (Watson 1956). The overall mean field has an inclination of 73.3°, which is almost 4° lower than the inclination of a central axial dipole field in northern Iceland. Some studies in the Tertiary lavas of Iceland have yielded similar results indicating "far-sided" and/or "right-handed" mean poles, while the mean fields from others are very close to the dipole field. See discussion in Section 3.5 of Kristjansson et al. (2003). We shall not attempt to account for these variations in terms of persistent geomagnetic non-dipole fields or lateral displacements of Iceland. Rather, they might for example be due to chance effects in the longitudinal distribution of mid- and low-latitude poles in the collections, or due to the presence of regional magnetic anomalies of crustal origin at the time of emplacement. Also, small systematic errors in the tectonic tilt corrections or in the orientation of samples may affect the overall results.

The angular standard deviation of virtual geomagnetic poles in the present collection has already been used by Kristjansson (2002) and Kristjansson et al. (2003) in a continuing study of possible long-term changes in this parameter. The data point which is marked as "4.2/3.2" in Fig. 3 of the former paper, agrees well with comparable data from large collections in other areas in Iceland.

Table 2 Average remanence directions and VGPs in the collection of Table 1 after rejection of six units (see text) and inversion of those where the VGPs have negative latitudes

Fields	N	R	Dec.	Inc.	k	a.s.d.	α_{95}
All lavas	312	288.6	359.9	73.4	13.3	22.3	2.3
Lavas with VGP lat.>45°	272	263.5	359.1	73.3	32.0	14.4	1.5
N lavas, VGP lat.>45°N	133	128.9	358.0	72.3	32.2	14.3	2.2
R lavas, VGP lat.>45°S	139	134.7	0.2	74.2	31.8	14.4	2.2
VGPs	N	R	Lon.	Lat.	k	a.s.d.	α_{95}
All lavas	312	268.2	164.1	86.1	7.1	30.7	3.2
Lavas with VGP lat. >45°	272	250.6	166.7	85.3	12.7	22.9	2.5

N=number of units averaged. R=length of the sum of N unit vectors. Dec, Inc=declination and inclination of the mean field. k=Fisher's (1953) precision parameter. a.s.d.= angular standard deviation, estimated here as arc cos ((R-1)/(N-1)). Correction for within-flow scatter will reduce the a.s.d. values by 0.1–0.2°. With reference to Table 1 of Cox (1969) and the assumption of some serial correlation, the 95% confidence limits on the a.s.d. values are about ±6%. α_{95}=95% confidence angle for the mean. Lon, Lat=longitude and latitude of the mean VGP

The average intensity of remanence (J100) in normally polarized lavas of the present collection is about 20% higher than that in the reversely polarized lavas. One possible explanation of this difference is that even minor regional hydrothermal alteration tends to reduce the intensity of the primary remanence (Watkins and Walker 1977, their Fig. 13). Relatively fresh lavas occur in some large profile segments (e.g. in HO), which happen to be of normal polarity, while others that are more altered (SG, lower part of KG, etc.) are predominantly of reverse polarity.

Conclusions

We have measured paleomagnetic directions in 319 lava flows, cored in eight mapped mountainside profiles through the lava pile in central Northern Iceland. When profile overlaps are left out (chiefly in profile KG), a composite section compiled from these contains about 270 flows of 2.9 km thickness. Its age range, estimated from preliminary Ar-Ar dating (Hardarson et al. 1999) in these profiles as well as from K-Ar dating and mapping results in nearby areas (see section on Geology above) is between 9 and 5 Ma ago. This section includes minor hyaloclastites and other volcanogenic sediments totaling around 200 m. Additionally, the sediments capping profile AS may reach 200 m; Johannesson (1991) assumes them to be deposited by erosion when activity in the Öxnadalur central volcano declined around 8 Ma ago.

The primary remanence in samples from these lava flows is very stable and directionally consistent within each unit. The number of polarity reversals in the composite section is 17, in addition to excursions usually recorded in one or two flows each (including the flows KG 55 and 58 where the VGP crosses the Equator). The scatter of VGP positions fits a temporal trend obtained from other large surveys in Iceland (Kristjansson 2002). In two groups of several flows at polarity zone boundaries, irregular fluctuations of the field are recorded (Fig. 5).

At present, our knowledge of the age of the composite section is insufficient to allow definite correlations of its polarity zones (Fig. 3) with particular zones in such sections elsewhere in Iceland, or with the geomagnetic polarity time scale. However, it seems likely that the normal-polarity zone occurring in our profile GL (Figs. 2, 3) correlates (through KF of Figs. 1a, 2) to the topmost normal zone of profile PF of Saemundsson et al. (1980), and that it corresponds to the geomagnetic chron C4An (8.7–9.0 Ma ago) of Cande and Kent (1995). Higher up in the composite section the correlation problem is compounded by the hiatus (and possible unconformity) represented by the thick sediments between our profiles AS and SG. Another hiatus may be indicated in profile AF where a polarity reversal between flows 14 and 15 coincides with a 17 m sediment layer. In contrast, there are numerous instances within the composite section where the interval between successive lavas was probably very short (<<1 ka, judging from clustered remanence directions), especially in profile HO.

The present study is in many ways only a reconnaissance effort in the stratigraphy of the valleys of Eyjafjördur, but it establishes a base for further stratigraphic mapping in this area and its surroundings. For instance, the thick sediments between profiles AS and SG (Figs. 2, 3) can be traced for at least 40 km (Johannesson 1991), and our work may be expected to aid in the correlation of sequences around these sediments in other localities. The relatively unaltered lava flows in this area may be suitable material for accurate radiometric age determinations of geomagnetic polarity transitions and of significant volcanic and climatic events, as has been demonstrated in Eastern Iceland by Duncan and Helgason (1998). The entire geological history of the Eyjafjördur-Skagafjördur area is still mostly uncharted, regarding for instance the activity of the local central-volcano complexes, tectonics, and erosion. On a larger scale, the general framework of spreading and plume activity in Iceland during the last 15 Ma is also a matter of considerable uncertainty.

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References

Aronson JL, Saemundsson K (1975) Relatively old basalts from structurally high areas in central Iceland. Earth Planet Sci Lett 28:83–97

Cande SC, Kent DV (1995) Revised calibration of the geomagnetic polarity timescale for the late Cretaceous and Cenozoic. J Geophys Res 100:6093–6095

Cox A (1969) Confidence limits for the precision parameter κ. Geophys J R Astr Soc 18:545–549

Duncan RA, Helgason J (1998) Precise dating of the Holmatindur cooling event in eastern Iceland: evidence for mid-Miocene bipolar glaciation. J Geophys Res 103:12,397–12,404

Einarsson T (1957) Der Paläomagnetismus der isländischen Basalte und seine stratigraphische Bedeutung. Neues Jb Geol Paläontol Monatsh 1957:159–175

Fisher RA (1953) Dispersion on a sphere. Proc R Soc Lond A 217:295–305

Goguitchaichvili A, Prévot M, Camps P (1999) No evidence for strong fields during the R3-N3 Icelandic geomagnetic reversal. Earth Planet Sci Lett 167:15–34

Hardarson BS, Fitton JG, Ellam RM, Pringle MS (1997) Rift relocation—a geochemical and geochronological investigation of a paleo-rift in northwest Iceland. Earth Planet Sci Lett 153:181–196

Hardarson BS, Pringle MS, Kristjansson L, Gudmundsson A, Johannesson H (1999) Stratigraphy and paleomagnetism of a 2.8 km lava succession in central northern Iceland. Paper GA1.15/E/03-B4 presented at the XXII General Assembly,

International Union of Geodesy and Geophysics, Birmingham. Abstracts Volume B:326
Hebeda EH, Hantelmann JJ, Priem HNA (1974) K-Ar and paleomagnetic investigations of a series of lava flows on the eastern side of Eyjafjördur, Northern Iceland. Paper presented at the International Meeting for Geochronology, Cosmochronology and Isotope Geology, Paris (mimeographed abstract, 3 pp)
Hjartarson A, Jonsdottir HE (1999) Akureyri—Geological Map. National Energy Authority of Iceland, Report OS-99118, pp 1–18 and color map in scale 1:50,000 (in Icelandic)
Hospers J (1954) Magnetic correlation in volcanic districts. Geol Mag 91:352–360
Johannesson H (1991) Yfirlit um jardfraedi Tröllaskaga (Overview of the geology of the Tröllaskagi peninsula, in Icelandic). Yearbook of the Iceland Touring Association 1991, pp 39–56
Johannesson H, Saemundsson K (1998) Geological Map of Iceland in scale 1:500,000. Tectonics. Icelandic Institute of Natural History, Reykjavik
Kristjansson L (1999) On low-latitude virtual geomagnetic poles in Icelandic basalt lava sequences. Phys Earth Planet Int 115:137–145
Kristjansson L (2002) Estimating properties of the paleomagnetic field from Icelandic lavas. Phys Chem Earth 27:1205–1213
Kristjansson L, Gudmundsson A (2001) Paleomagnetic studies in Skardsheidi, South-Western Iceland. Jökull 50:33–48
Kristjansson L, Johannesson H (1996) Stratigraphy and paleomagnetism of the lava pile south of Isafjardardjup, NW-Iceland. Jökull 44:3–16
Kristjansson L, Johannesson H (1999) Secular variation and reversals in a composite 2.5 km thick lava section in central Western Iceland. Earth Planets Space 51:261–276
Kristjansson L, Sigurgeirsson M (1993) The R3-N3 and R5-N5 paleomagnetic transition zones in Iceland revisited. J Geomag Geoel 45:275–288
Kristjansson L, Fridleifsson IB, Watkins ND (1980) Stratigraphy and paleomagnetism of the Esja, Eyrarfjall and Akrafjall mountains, SW-Iceland. J Geophys 47:31–42
Kristjansson L, Gudmundsson A, Haraldsson H (1995) Stratigraphy and paleomagnetism of a 3-km thick Miocene lava pile in the Mjoifjördur area, eastern Iceland. Geol Rundsch 84:813–830
Kristjansson L, Hardarson BS, Audunsson H (2003) A detailed paleomagnetic study of the oldest (15 m.y.) lava sequences in Northwest Iceland. Geophys J Int 155:991–1015
Mankinen EA, Prévot M, Grommé CS, Coe RS (1985) The Steens Mountain (Oregon) geomagnetic polarity transition, 1. Directional history, duration of episodes, and rock magnetism. J Geophys Res 90:10,393–10,416
McDougall I, Saemundsson K, Johannesson H, Watkins ND, Kristjansson L (1977) Extension of the geomagnetic polarity time scale to 6.5 m.y.: K-Ar dating, geological and paleomagnetic study of a 3,500-m lava succession in western Iceland. Bull Geol Soc Am 88:1–15
McDougall I, Kristjansson L, Saemundsson K (1984) Magnetostratigraphy and geochronology of Northwest Iceland. J Geophys Res 89:7029–7060
McElhinny MW, McFadden PL, Merrill RT (1996) The myth of the Pacific dipole window. Earth Planet Sci Lett 143:13–22
Saemundsson K, Kristjansson L, McDougall I, Watkins ND (1980) K-Ar dating, geological and paleomagnetic study of a 5-km lava succession in Northern Iceland. J Geophys Res 85:3628–3646
Thoroddsen T (1901) Geological Map of Iceland, in scale 1:600,000. The Carlsberg Fund, Copenhagen
Walker GPL (1959) Geology of the Reydarfjördur area, Eastern Iceland. Quart J Geol Soc Lond 114:367–393
Watkins ND, Walker GPL (1977) Magnetostratigraphy of Eastern Iceland. Am J Sci 277:513–584
Watson GS (1956) Analysis of dispersion on a sphere. Monthly Not R Astr Soc Geophys Suppl 7:153–159